The ξ–integral, run in Wolfram

$\int_{\Lambda/k}^{1-\Lambda/k} dz\, \left[z\bar z + \frac{z}{\bar z} + \frac{\bar z}{z}\right] \log\dfrac{a}{1-z}$ — raw output, reduction, limit

$a = (y-x)^2/m^2$, $\lambda = \Lambda/k$, $\ell_k = \log(k/\Lambda)$ — every line below is verbatim kernel output

0. Result

$$\int_{\Lambda/k}^{1-\Lambda/k} dz\, \left[z\bar z + \frac{z}{\bar z} + \frac{\bar z}{z}\right] \log\frac{a}{1-z} \;=\; \Big(2\ell_k - \frac{11}{6}\Big)\log a + \frac{\ell_k^2}{2} + \frac{\pi^2}{6} - \frac{67}{36} \;+\; \frac{\Lambda}{k}\Big(2\log a + \ell_k\Big) + O\big(\lambda^2\log\lambda\big).$$

Both the limit and the first correction are produced by the kernel, not by hand (§3, §4). Note $z\bar z + \frac{z}{\bar z} + \frac{\bar z}{z} = P_{gg}(z)/2N_c$ and $-\frac{11}{6} \times 2N_c = -b$ at $N_f = 0$.

1. What was run

Giving the kernel the range assumption avoids the `ConditionalExpression` wrapper — the expression inside is identical either way.

```
res = Integrate[Log[a/(1 - z)]*(z (1 - z) + z/(1 - z) + (1 - z)/z),
   {z, lam, 1 - lam}, Assumptions -> {0 < lam < 1/2, a > 0}];
InputForm[Simplify[res, 0 < lam < 1/2 && a > 0]]
```

Raw output

```
(-67 + 138*lam - 12*lam^2 + 8*lam^3 - 6*Log[1 - lam]
 - 72*Log[-(a/(-1 + lam))] + 72*lam*Log[-(a/(-1 + lam))]
 - 18*lam^2*Log[-(a/(-1 + lam))] + 12*lam^3*Log[-(a/(-1 + lam))]
 - 18*Log[-(a/(-1 + lam))]^2
 + 6*Log[a/lam] + 72*lam*Log[a/lam] - 18*lam^2*Log[a/lam] + 12*lam^3*Log[a/lam]
 + 36*Log[1 - lam]*Log[a/lam] + 18*Log[a/lam]^2
 + 6*Log[lam] - 36*Log[-(a/(-1 + lam))]*Log[lam]
 + 36*PolyLog[2, 1 - lam] - 36*PolyLog[2, lam])/36
```

$\mathrm{Log}[-(a/(-1+\lambda))] = \log\frac{a}{1-\lambda}$; this is the same expression as the `ConditionalExpression`, with k scaled out ($\lambda = \Lambda/k$).

2. Reduction — collect in $\log a$

Rewrite every logarithm in $A = \log a$, $\ell = -\log\lambda$, $\omega = \log(1-\lambda)$ and collect:

```
r = res /. {Log[-(a/(-1 + lam))] -> A - w, Log[a/lam] -> A + l,
            Log[1 - lam] -> w, Log[lam] -> -l};
r  = Collect[Expand[r], A, Simplify];
cA = Simplify[Coefficient[r, A, 1]];
rest = Simplify[r - cA*A];
```

Output

```
coefficient of Log[a] :
   -11/6 + 2*l + 4*lam - lam^2 + (2*lam^3)/3 + 2*w

remainder :
   (-67 + 18*l^2 + 138*lam + 72*l*lam - 12*lam^2 - 18*l*lam^2 + 8*lam^3
    + 12*l*lam^3 + 66*w - 72*lam*w + 18*lam^2*w - 12*lam^3*w - 18*w^2
    + 36*PolyLog[2, 1 - lam] - 36*PolyLog[2, lam])/36
```

So, exactly, at any Λ/k:

$$I = \mathcal{A}(\lambda)\log a + \mathcal{B}(\lambda), \qquad \mathcal{A}(\lambda) = 2\ell + 2\omega - \frac{11}{6} + 4\lambda - \lambda^2 + \frac{2}{3}\lambda^3,$$
$$\mathcal{B}(\lambda) = \frac{1}{36}\Big[18\ell^2 - 67 + 36\big(\mathrm{Li}_2(1-\lambda) - \mathrm{Li}_2(\lambda)\big) + 66\,\omega - 18\,\omega^2$$
$$+138\lambda - 12\lambda^2 + 8\lambda^3 + 6(\ell-\omega)\big(12\lambda - 3\lambda^2 + 2\lambda^3\big)\Big].$$

The λ–dependent part of the remainder is exactly $6(\ell-\omega)(12\lambda-3\lambda^2+2\lambda^3)$: the kernel's $72\ell\lambda - 18\ell\lambda^2 + 12\ell\lambda^3$ and $-72\lambda\omega + 18\lambda^2\omega - 12\lambda^3\omega$ are its two halves. The $-\frac{11}{6}$ in $\mathcal{A}$ is the constant part of $\frac{1}{6}(Q-P)$, $Q-P = -11+24\lambda-6\lambda^2+4\lambda^3$, the mismatch between the two polynomial brackets of the raw output.

3. The limit $\Lambda/k \to 0$

Li_2 has a branch point at 1, so `Series` will not expand $\mathrm{PolyLog}[2, 1-\mathrm{lam}]$ directly — substitute the inversion identity $\mathrm{Li}_2(1-x) = \frac{\pi^2}{6} - \log x \log(1-x) - \mathrm{Li}_2(x)$ first. And Mathematica will not split $\mathrm{Log}[\mathrm{a}/\mathrm{lam}]$ without sign information, so supply that rule too, otherwise the answer *looks* different from the target and a `Simplify[... ] === 0` test returns `False` even though the two agree.

```
r = res /. PolyLog[2, 1 - lam] :>
             Pi^2/6 - Log[lam] Log[1 - lam] - PolyLog[2, lam];
split = {Log[a/lam] -> Log[a] - Log[lam],
         Log[-(a/(-1 + lam))] -> Log[a] - Log[1 - lam]};

lim  = Simplify[Expand[Normal[Series[r, {lam, 0, 0}]] /. split]];
corr = Simplify[Expand[(Normal[Series[r, {lam, 0, 1}]]
                        - Normal[Series[r, {lam, 0, 0}]]) /. split]];
lk = -Log[lam];

Collect[lim, {Log[a], Log[lam]}, Simplify]
Simplify[lim - ((2 lk - 11/6) Log[a] + lk^2/2 + Pi^2/6 - 67/36)] === 0
corr
Simplify[corr - lam (2 Log[a] + lk)] === 0
```

Output

```
LIMIT      : (-67 + 6*Pi^2)/36 + Log[a]*(-11/6 - 2*Log[lam]) + Log[lam]^2/2
  == (2 lk - 11/6) Log a + lk^2/2 + Pi^2/6 - 67/36 ?   True

CORRECTION : lam*(2*Log[a] - Log[lam])
  == lam (2 Log a + lk) ?                              True
```

Reading the first line with $-\log\lambda = \ell_k$:

$$I\big|_{\lambda\to 0} = \Big(2\ell_k - \frac{11}{6}\Big)\log a + \frac{\ell_k^2}{2} + \frac{\pi^2}{6} - \frac{67}{36}.$$

Provenance of the four numbers, as they appear in the output: $2\ell_k$ and $-\frac{11}{6}$ are the $\lambda\to 0$ value of $\mathcal{A}$; $\frac{\ell_k^2}{2}$ is $18\ell^2/36$; $\frac{\pi^2}{6}$ is $6\pi^2/36$, i.e. $36\,\mathrm{Li}_2(1) = 6\pi^2$; and $-\frac{67}{36}$ is the -67 that was in the raw output from the start.

4. What the limit discards

The kernel's second line: the whole $O(\Lambda/k)$ correction is

$$I(a,\lambda) - I(a,0) = \frac{\Lambda}{k}\Big[2\log a + \ell_k\Big] + O\big(\lambda^2\log\lambda\big).$$

Numerically, at $a = 100$ this is 1.8×10^{-3} at $\lambda = 10^{-4}$ and 2.8×10^{-7} at $\lambda = 10^{-8}$, against an answer of order 10^2. Since Λ/k^+ is the rapidity cutoff and is sent to zero throughout (cf. the remark before Eq. (C.5) of arXiv:1610.03453), these terms sit far below the accuracy of everything else in the calculation.

5. The whole session, copy–pasteable

```
res = Integrate[Log[a/(1 - z)]*(z (1 - z) + z/(1 - z) + (1 - z)/z),
   {z, lam, 1 - lam}, Assumptions -> {0 < lam < 1/2, a > 0}];

(* exact, collected in Log[a] *)
r0 = res /. {Log[-(a/(-1 + lam))] -> A - w, Log[a/lam] -> A + l,
             Log[1 - lam] -> w, Log[lam] -> -l};
r0 = Collect[Expand[r0], A, Simplify];
cA = Simplify[Coefficient[r0, A, 1]]
rest = Simplify[r0 - cA*A]

(* the limit *)
r = res /. PolyLog[2, 1 - lam] :>
             Pi^2/6 - Log[lam] Log[1 - lam] - PolyLog[2, lam];
split = {Log[a/lam] -> Log[a] - Log[lam],
         Log[-(a/(-1 + lam))] -> Log[a] - Log[1 - lam]};
lk = -Log[lam];

lim = Simplify[Expand[Normal[Series[r, {lam, 0, 0}]] /. split]];
Collect[lim, {Log[a], Log[lam]}, Simplify]
Simplify[lim - ((2 lk - 11/6) Log[a] + lk^2/2 + Pi^2/6 - 67/36)] === 0

corr = Simplify[Expand[(Normal[Series[r, {lam, 0, 1}]]
                        - Normal[Series[r, {lam, 0, 0}]]) /. split]]
Simplify[corr - lam (2 Log[a] + lk)] === 0

(* numerical spot check *)
NIntegrate[Log[100/(1 - z)] (z (1 - z) + z/(1 - z) + (1 - z)/z),
   {z, 10^-6, 1 - 10^-6}, WorkingPrecision -> 30]
(* 214.0207546140127...  vs the lam->0 formula 214.020731588185... *)
```

Two traps, both of which make the answer look wrong when it is not. (i) `Series` silently refuses to expand `PolyLog[2, 1 - lam]` about $\lambda = 0$ — the branch point is there — so the $\pi^2/6$ never appears unless you substitute the inversion identity. (ii) `Simplify` will not turn `Log[a/lam]` into `Log[a] - Log[lam]` without sign assumptions, so two equal expressions can fail an `=== 0` test.